AI SURVEILLANCE SYSTEM

CCTV Capacity & Specification Report

This system provides automated, AI-driven monitoring of CCTV camera feeds - purpose-built for exam-centre and high-security surveillance. It captures frames from the camera streams and analyses them to understand each scene and detect violations in near real time - with snapshot evidence on a central dashboard.

1,00,000	20	~2.8 TB	24x7
Cameras supported	NVIDIA H200 GPUs	Total GPU memory	Automated operation

Hardware & Compute Capacity

The system runs on data-centre-class NVIDIA H200 GPUs across two server types: four single-GPU servers and two high-density servers that each carry eight H200 GPUs and 2 TB of system RAM.

Resource	Specification
GPU type	NVIDIA H200 NVL - 141 GB per GPU
Total GPUs	20 (~2,820 GB / ~2.8 TB total GPU memory)
Single-GPU servers	4 servers x 1 H200, 128 GB RAM each
High-density servers	2 servers x 8 H200 (~1.15 TB VRAM each), 2 TB RAM each
Total system RAM	~4.5 TB (4 x 128 GB + 2 x 2 TB)

Key Specifications

Capability	Current Standing
Cameras supported	Up to 1,00,000 (1 lakh) cameras
Camera link types	RTSP and HLS
Processing	Every camera classified, then monitored for violations
Violation types	15+ (phone, crowd, intruder, proximity, etc.)
Evidence	Snapshot + timestamp logged for every detected incident
Availability	Continuous 24x7 automated operation

How the System Works - Per-Camera Flow

Every camera follows the same two-step journey. The system first classifies the camera to understand what it shows, then runs violation detection - but only on cameras that are successfully classified and accessible.

Step 1 - Scene Classification

1. **Check accessibility.** The system first connects to the camera and confirms it is reachable. Cameras that are offline or unreachable are marked inaccessible and skipped.
2. **Capture a frame.** For an accessible camera, a single frame is captured from the stream.
3. **Classify the scene.** The frame is sent to a Vision-Language AI model (Qwen3-VL) that identifies the scene - e.g. Computer Lab, Server Room, Entry or Parking Lot. Frames that are dark, blank or unclear are labelled "Uncertain / Obstructed"; all others are assigned their scene category.

Step 2 - Violation Detection (on accessible, classified cameras)

4. **Detect.** At a set interval the system captures a fresh frame and runs high-speed object-detection models (YOLO), matched to the camera's scene, to look for violations - people, phones, crowding, movement in restricted zones, and more.
5. **Act.** If a violation is found, the system saves a snapshot as evidence, logs the incident (camera, type and time), and surfaces it on the operator dashboard. The cycle repeats continuously, so violations appear in near real time.

```
Check accessibility  -->  Capture frame  -->  Classify (vLLM)
      Capture fresh frame  -->  YOLO detection  -->  Violation?  -->  Snapshot + log + alert
```

Violations & Events Detected

Phone phone usage in monitored areas	**Server Room** people or movement in server rooms
Crowd overcrowding at entry/exit & biometric	**Entry/Exit** movement patterns at doorways
Low/Zero Occupancy empty or understaffed rooms	**Invigilator** invigilator presence & movement
Proximity candidates sitting too close	**Tampering** blur, blockage or feed interference
Intruder unauthorised persons in protected areas	**Inversion** rotated or wrongly mounted cameras
Strong Room movement inside secured rooms	**Blind Spot** partial visibility or frozen feeds
QP Box question-paper box presence/absence	